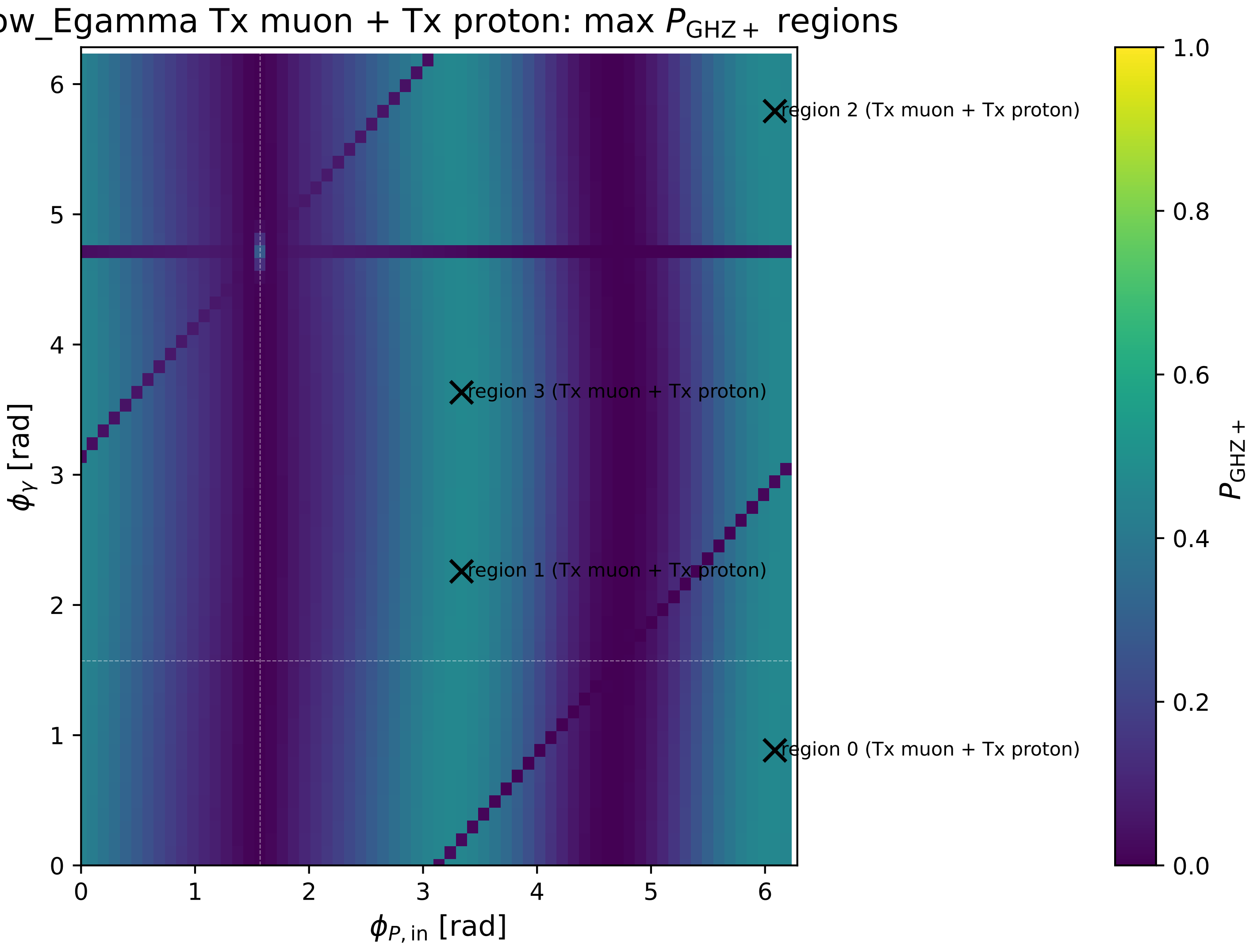
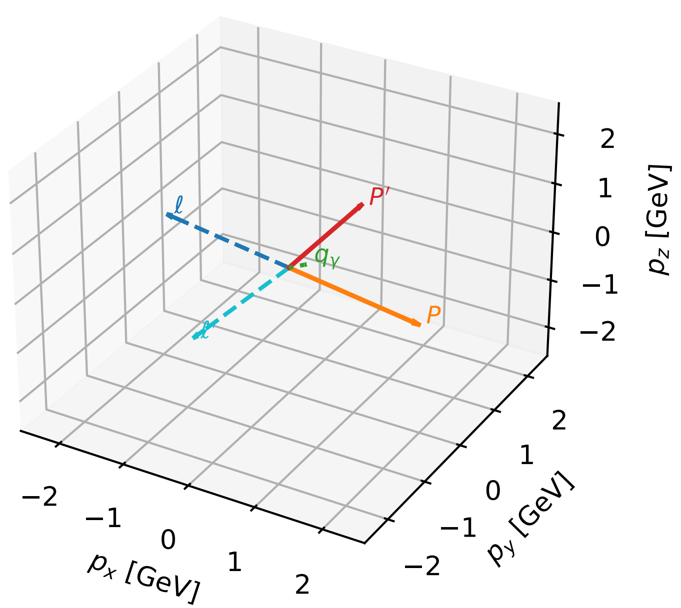

ow_Egamma Tx muon + Tx proton: max $P_{\mathrm{GHZ}+}$ regions
region 2 (Tx muon + Tx proton)
region 3 (Tx muon + Tx proton)
region 1 (Tx muon + Tx proton)
region 0 (Tx muon + Tx proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$P_{\mathrm{GHZ}+}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

Tx muon + Tx proton characteristic max $P_{\mathrm{GHZ+}}$ configuration

3D momenta

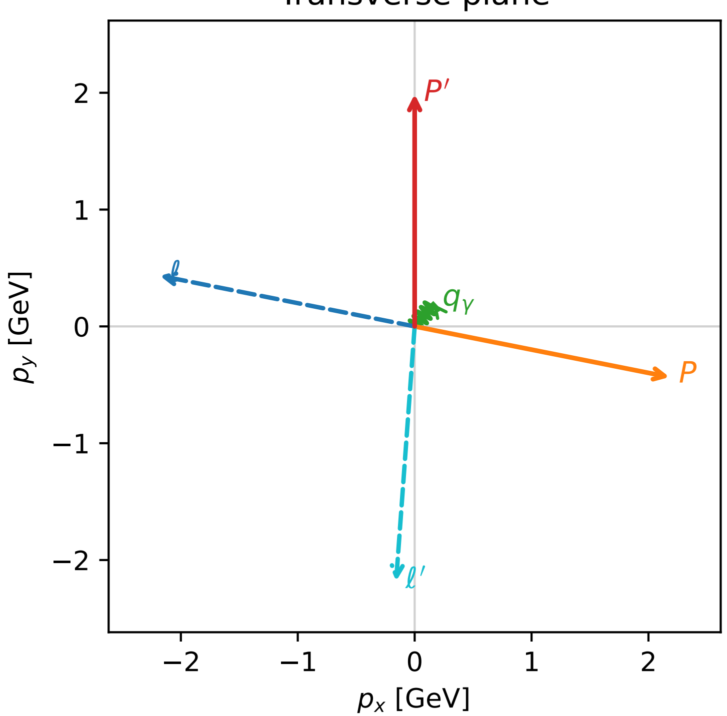


ghz_purity_low_egamma_txtx_region_0 P_{\rm GHZ+}=0.465057
region: low_Egamma_TxTx_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$

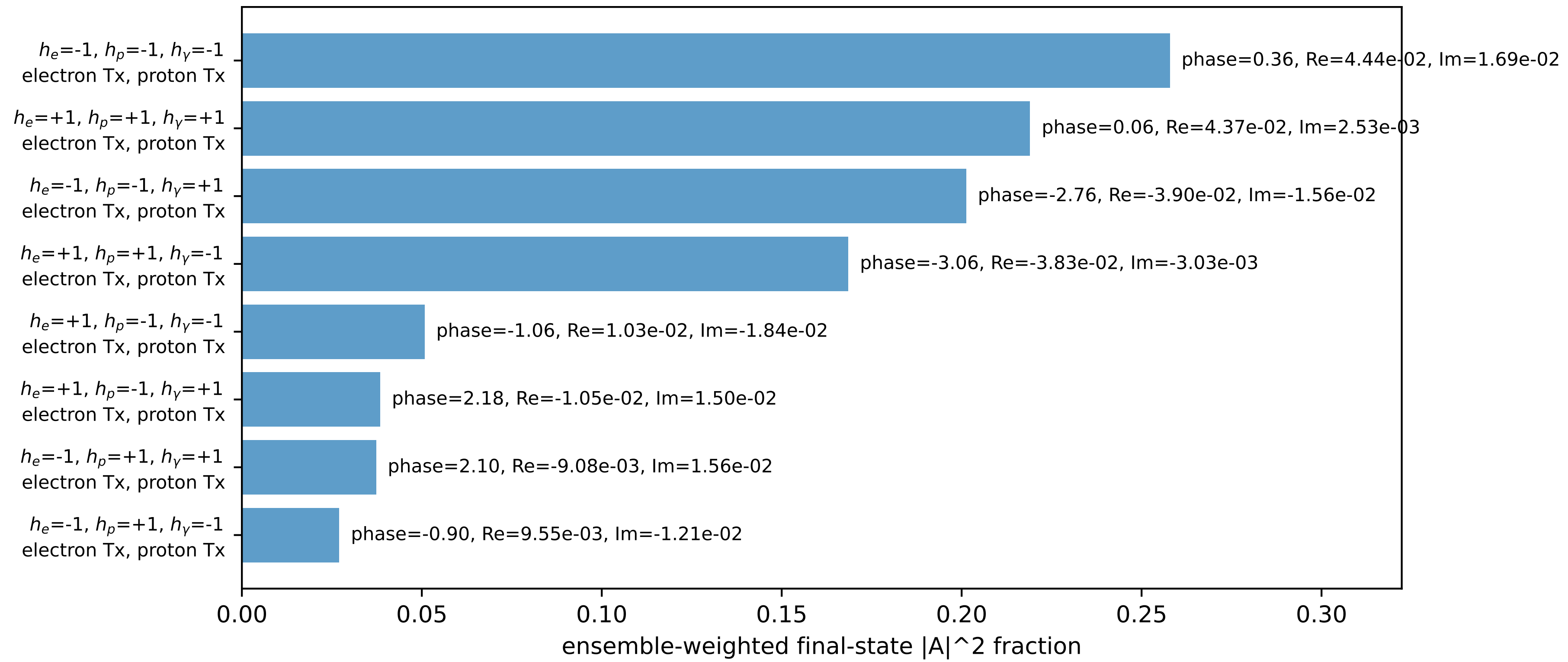
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.98841
θ_{in}=1.57079, ϕ_ℓ=2.94524, ϕ_P=6.08684
E_γ=0.25, ϕ_γ=0.883573
Q^2=10.9265, x_B=0.97062, t=-10.5748, W^2=1.21059, y=0.545812
$\theta(\ell', \gamma)$=144.783 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -2.1804 p_y= 0.4337 p_z= -0.0000
P E= 2.4129 p_x= 2.1804 p_y= -0.4337 p_z= 0.0000
ℓ' E= 2.1900 p_x= -0.1586 p_y= -2.1817 p_z= 0.0000
P' E= 2.1985 p_x= 0.0000 p_y= 1.9884 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.1586 p_y= 0.1933 p_z= 0.0000

Transverse plane

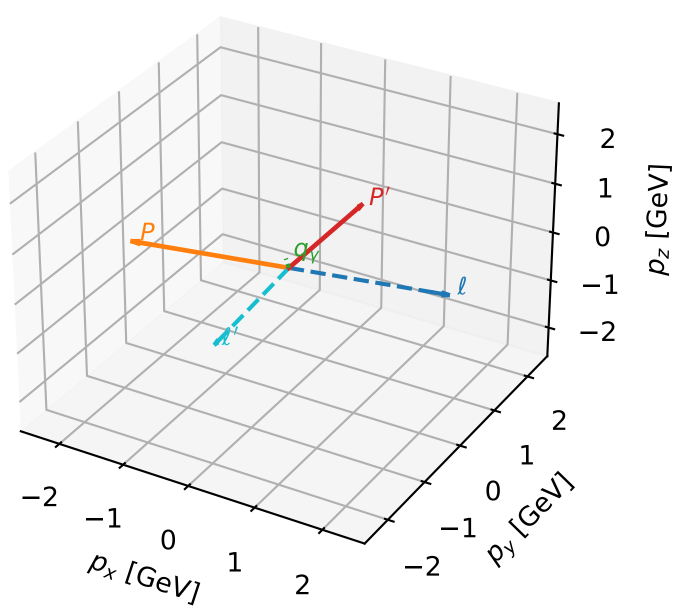


Leading initial-ensemble/final-state components (N=8, retained=100.0%)



Tx muon + Tx proton characteristic max $P_{\rm GHZ+}$ configuration

3D momenta

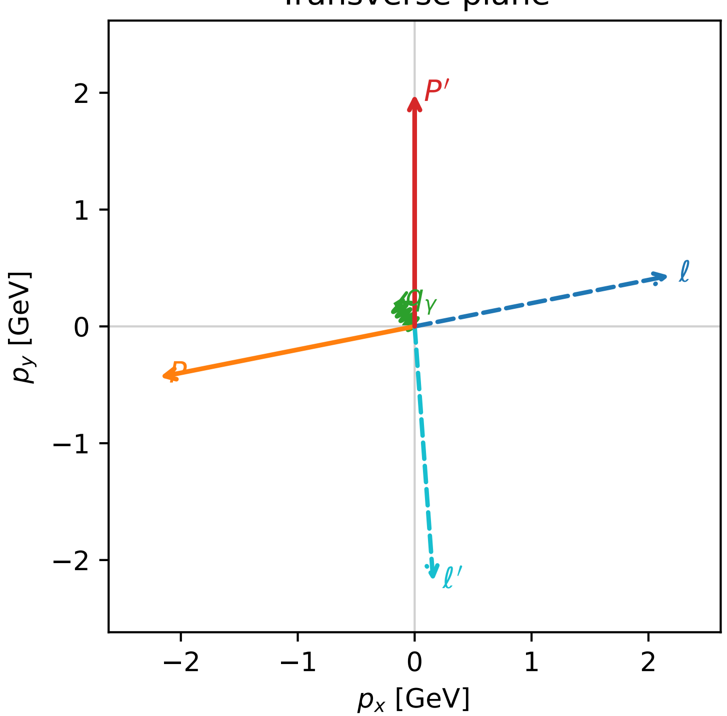



```
ghz_purity_low_egamma_txtx_region_1  P_{\rm GHZ+}=0.465057
region: low_Egamma_TxTx_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
```
incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$

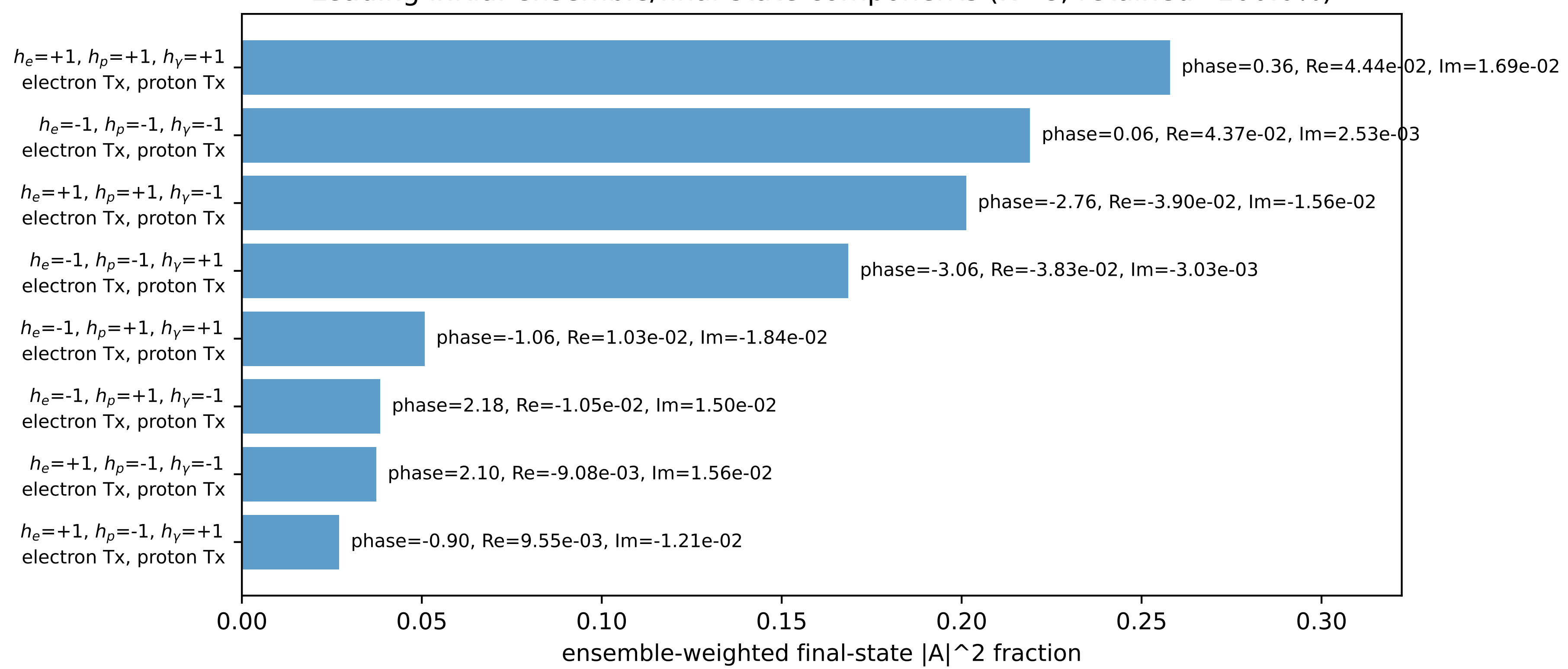
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.98841
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0.19635, ϕ_P=3.33794
E_γ=0.25, ϕ_γ=2.25802
Q^2=10.9265, x_B=0.97062, t=-10.5748, W^2=1.21059, y=0.545812
$\theta(\ell', \gamma)$=144.783 deg

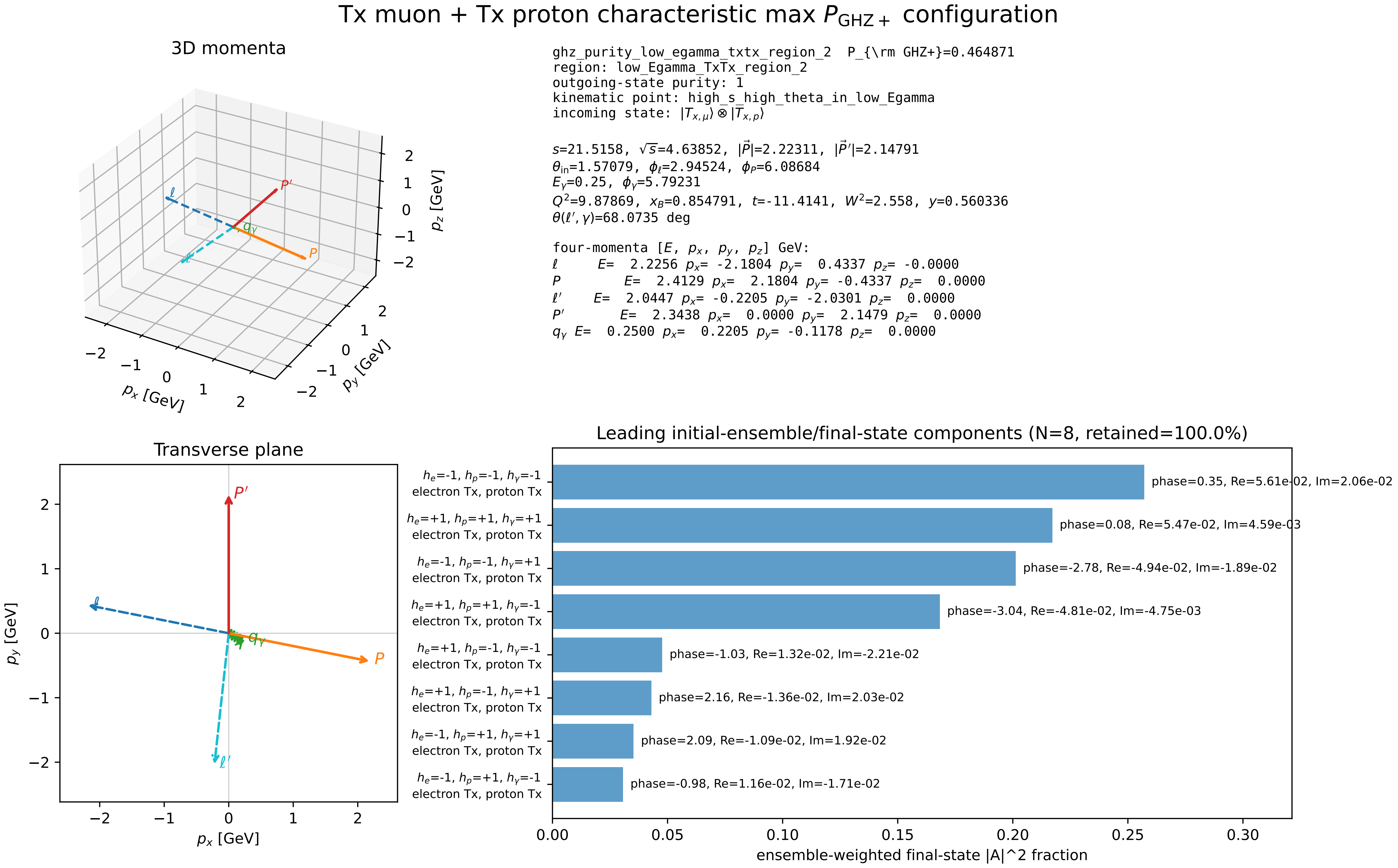
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 2.1804 p_y= 0.4337 p_z= -0.0000
P E= 2.4129 p_x= -2.1804 p_y= -0.4337 p_z= 0.0000
ℓ' E= 2.1900 p_x= 0.1586 p_y= -2.1817 p_z= 0.0000
P' E= 2.1985 p_x= 0.0000 p_y= 1.9884 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.1586 p_y= 0.1933 p_z= 0.0000

Transverse plane

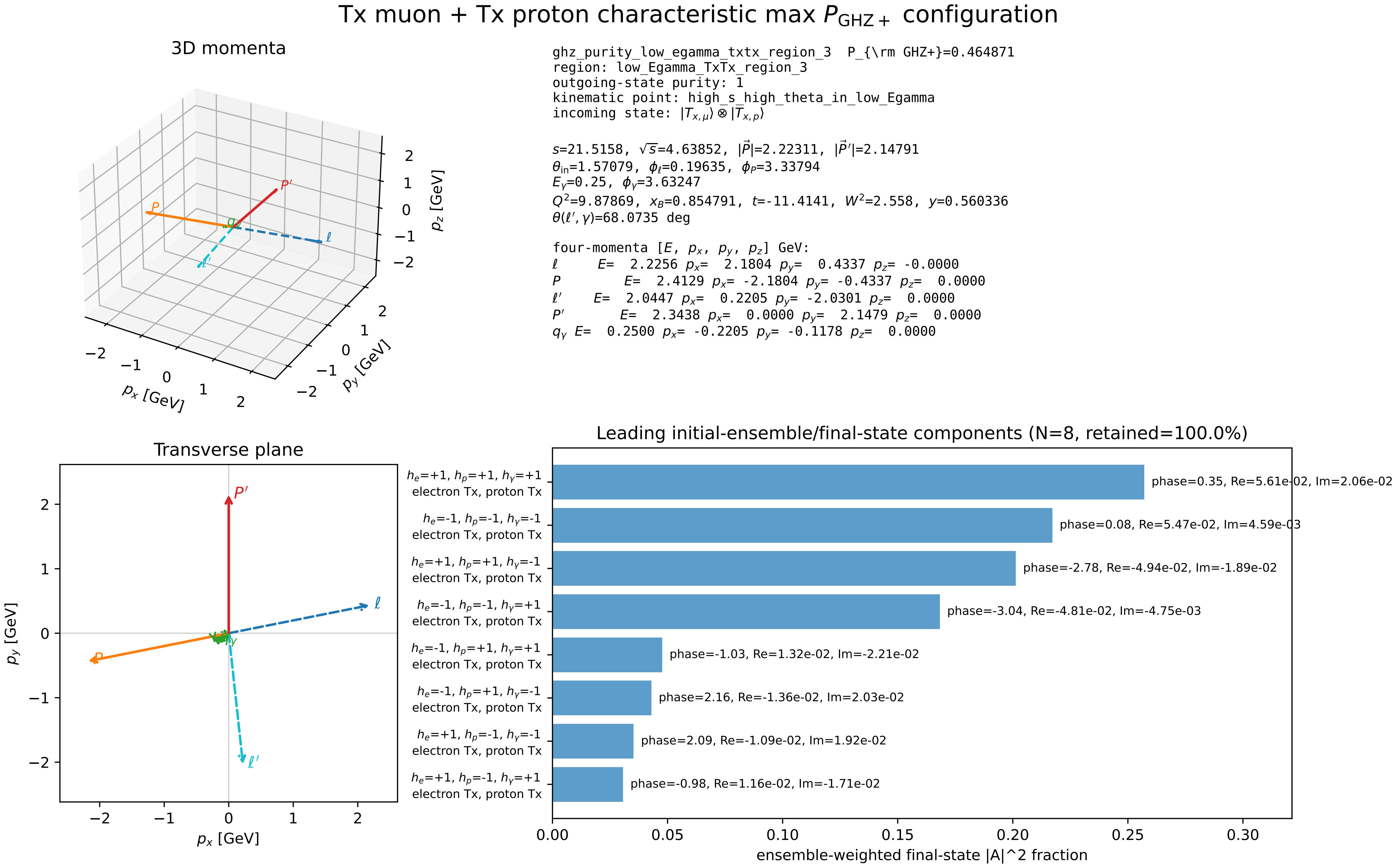


Leading initial-ensemble/final-state components (N=8, retained=100.0%)





Tx muon + Tx proton characteristic max $P_{\rm GHZ+}$ configuration
3D momenta
ℓ
P'
q_γ
P
ℓ'
p_x [GeV]
p_y [GeV]
p_z [GeV]
ghz_purity_low_egamma_txtx_region_2 P_{\rm GHZ+}=0.464871
region: low_Egamma_TxTx_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.14791
$\theta_{\rm in}$=1.57079, ϕ_ℓ=2.94524, ϕ_P=6.08684
E_γ=0.25, ϕ_γ=5.79231
Q^2=9.87869, x_B=0.854791, t=-11.4141, W^2=2.558, y=0.560336
$\theta(\ell',\gamma)$=68.0735 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -2.1804 p_y= 0.4337 p_z= -0.0000
P E= 2.4129 p_x= 2.1804 p_y= -0.4337 p_z= 0.0000
ℓ' E= 2.0447 p_x= -0.2205 p_y= -2.0301 p_z= 0.0000
P' E= 2.3438 p_x= 0.0000 p_y= 2.1479 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.2205 p_y= -0.1178 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=8, retained=100.0%)
h_e=-1, h_p=-1, h_γ=-1 electron Tx, proton Tx — phase=0.35, Re=5.61e-02, Im=2.06e-02
h_e=+1, h_p=+1, h_γ=+1 electron Tx, proton Tx — phase=0.08, Re=5.47e-02, Im=4.59e-03
h_e=-1, h_p=-1, h_γ=+1 electron Tx, proton Tx — phase=-2.78, Re=-4.94e-02, Im=-1.89e-02
h_e=+1, h_p=+1, h_γ=-1 electron Tx, proton Tx — phase=-3.04, Re=-4.81e-02, Im=-4.75e-03
h_e=+1, h_p=-1, h_γ=-1 electron Tx, proton Tx — phase=-1.03, Re=1.32e-02, Im=-2.21e-02
h_e=+1, h_p=-1, h_γ=+1 electron Tx, proton Tx — phase=2.16, Re=-1.36e-02, Im=2.03e-02
h_e=-1, h_p=+1, h_γ=+1 electron Tx, proton Tx — phase=2.09, Re=-1.09e-02, Im=1.92e-02
h_e=-1, h_p=+1, h_γ=-1 electron Tx, proton Tx — phase=-0.98, Re=1.16e-02, Im=-1.71e-02
ensemble-weighted final-state |A|^2 fraction


Tx muon + Tx proton characteristic max $P_{\rm GHZ+}$ configuration
3D momenta
P
P'
q_γ
ℓ
ℓ'
p_x [GeV]
p_y [GeV]
p_z [GeV]
ghz_purity_low_egamma_txtx_region_3 P_{\rm GHZ+}=0.464871
region: low_Egamma_TxTx_region_3
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.14791
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0.19635, ϕ_P=3.33794
E_γ=0.25, ϕ_γ=3.63247
Q^2=9.87869, x_B=0.854791, t=-11.4141, W^2=2.558, y=0.560336
$\theta(\ell', \gamma)$=68.0735 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 2.1804 p_y= 0.4337 p_z= -0.0000
P E= 2.4129 p_x= -2.1804 p_y= -0.4337 p_z= 0.0000
ℓ' E= 2.0447 p_x= 0.2205 p_y= -2.0301 p_z= 0.0000
P' E= 2.3438 p_x= 0.0000 p_y= 2.1479 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.2205 p_y= -0.1178 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=8, retained=100.0%)
h_e=+1, h_p=+1, h_γ=+1 electron Tx, proton Tx — phase=0.35, Re=5.61e-02, Im=2.06e-02
h_e=-1, h_p=-1, h_γ=-1 electron Tx, proton Tx — phase=0.08, Re=5.47e-02, Im=4.59e-03
h_e=+1, h_p=+1, h_γ=-1 electron Tx, proton Tx — phase=-2.78, Re=-4.94e-02, Im=-1.89e-02
h_e=-1, h_p=-1, h_γ=+1 electron Tx, proton Tx — phase=-3.04, Re=-4.81e-02, Im=-4.75e-03
h_e=-1, h_p=+1, h_γ=+1 electron Tx, proton Tx — phase=-1.03, Re=1.32e-02, Im=-2.21e-02
h_e=-1, h_p=+1, h_γ=-1 electron Tx, proton Tx — phase=2.16, Re=-1.36e-02, Im=2.03e-02
h_e=+1, h_p=-1, h_γ=-1 electron Tx, proton Tx — phase=2.09, Re=-1.09e-02, Im=1.92e-02
h_e=+1, h_p=-1, h_γ=+1 electron Tx, proton Tx — phase=-0.98, Re=1.16e-02, Im=-1.71e-02
ensemble-weighted final-state |A|^2 fraction